Lecture 18: Matroids and the Greedy Algorithm

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601.433/633 Introduction to Algorithms
Slides by Mike Dinitz

Introduction

Last time: somewhat greedy algorithm (Prim's), extremely greedy algorithm (Kruskal's)

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- Universe U
- Collection $\mathcal{I} \subseteq 2^U$ (so $I \subseteq U$ for all $I \in \mathcal{I}$). Called *independent sets*
- Weights $w : U \to \mathbb{R}_{\geq 0}$

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Problem: find *max weight* independent set

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So under weights w', max-weight IS = max-weight forest = max-weight spanning tree = min-weight spanning tree (weights w)

- So finding max-weight forest = finding min spanning tree.

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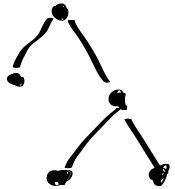
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Contradiction. □

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Definition

$(U, \mathcal{I})$ is a *matroid* if the following three properties hold:

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Warmup: In any matroid, the maximal independent sets (called bases) have the same size (called the rank of the matroid).

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  - $\varnothing$ linearly independent
  - If $\boldsymbol{F}$ linearly independent and $\boldsymbol{F}' \subseteq \boldsymbol{F}$, then $\boldsymbol{F}'$ linearly independent
  - Augmentation: if $\boldsymbol{F_1}$ linearly independent, $\boldsymbol{F_2}$ linearly independent, and $|\boldsymbol{F_2}| > |\boldsymbol{F_1}| \implies \dim(\text{span}(\boldsymbol{F_1})) = |\boldsymbol{F_1}| < |\boldsymbol{F_2}| = \dim(\text{span}(\boldsymbol{F_2}))$

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  - $\mathcal{I} = \{F \subseteq U : F \text{ linearly independent}\}$
  - $\varnothing$ linearly independent
  - If F linearly independent and $F' \subseteq F$, then F' linearly independent
  - Augmentation: if F_1 linearly independent, F_2 linearly independent, and $|F_2| > |F_1| \implies \dim(\text{span}(F_1)) = |F_1| < |F_2| = \dim(\text{span}(F_2))$

Matroids: generalize both graph theory and linear algebra!

- Originally invented by Whitney as an attempt to generalize the concept of "linear independence"

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We'll assume we have independence oracle.

Greedy Algorithm

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$F = \varnothing$
Sort U by weight (largest to smallest)
For each $u \in U$ in sorted order {
    If $F \cup \{u\} \in \mathcal{I}$, add u to F
}
Return F

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Contradiction! Greedy would add e_z next, not f_j.

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So for hereditary set systems, matroids exactly characterize when the greedy algorithm works!

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Contradiction. Suppose false $\implies$ $(U, \mathcal{I})$ hereditary but not matroid.

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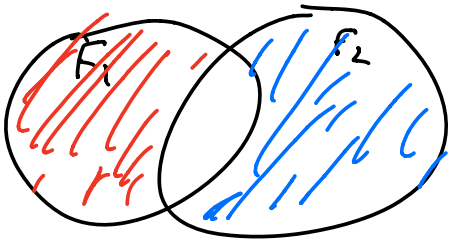
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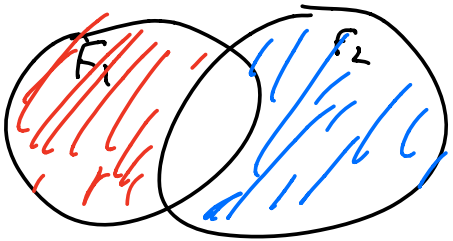
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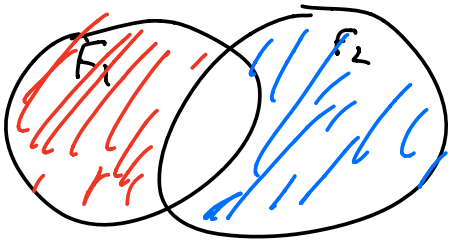
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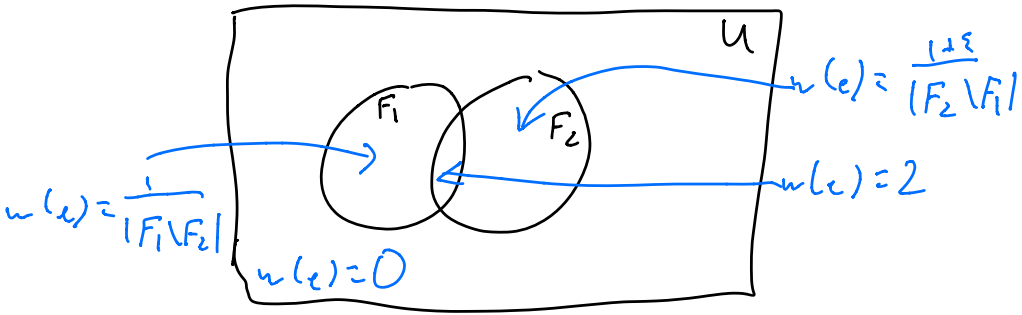
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$\implies \exists \epsilon > 0$ such that $0 < (1+\epsilon)|F_1 \setminus F_2| < |F_2 \setminus F_1|$

$$\implies \frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$$

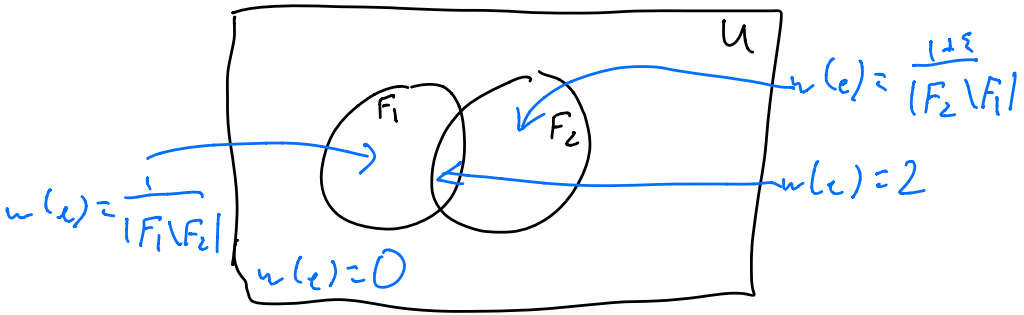
Proof (cont'd)

Use fact that $\frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$ to define weights.



Proof (cont'd)

Use fact that $\frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$ to define weights.

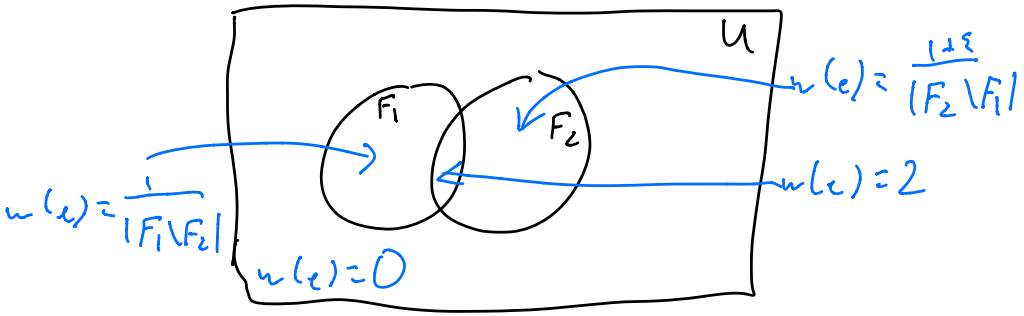


Greedy:

- Adds all of $F_1 \cap F_2$
- Adds all of $F_1 \setminus F_2$
- Can't add any of $F_2 \setminus F_1$

Proof (cont'd)

Use fact that $\frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$ to define weights.



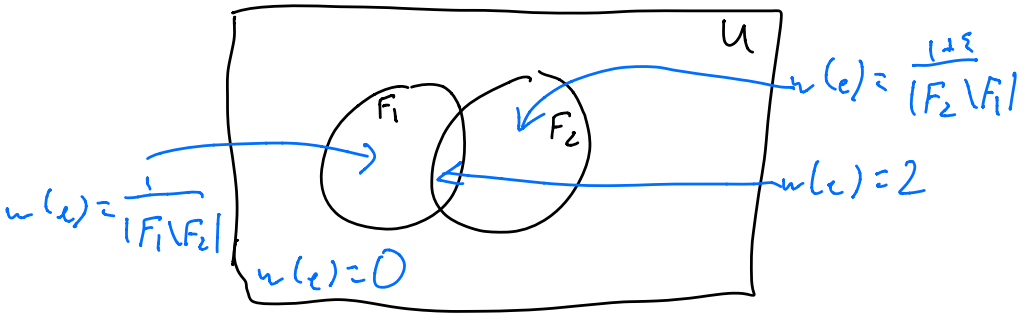
Greedy:

- Adds all of $F_1 \cap F_2$
- Adds all of $F_1 \setminus F_2$
- Can't add any of $F_2 \setminus F_1$

$$w(\text{greedy}) = 2|F_1 \cap F_2| + |F_1 \setminus F_2| \frac{1}{|F_1 \setminus F_2|}$$
$$= 2|F_1 \cap F_2| + 1$$

Proof (cont'd)

Use fact that $\frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$ to define weights.



Greedy:

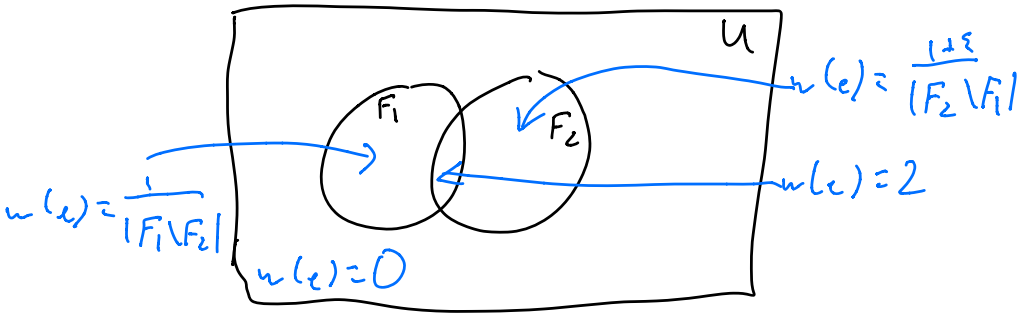
- Adds all of $F_1 \cap F_2$
- Adds all of $F_1 \setminus F_2$
- Can't add any of $F_2 \setminus F_1$

$$w(\text{greedy}) = 2|F_1 \cap F_2| + |F_1 \setminus F_2| \frac{1}{|F_1 \setminus F_2|}$$
$$= 2|F_1 \cap F_2| + 1$$

$$w(F_2) = 2|F_1 \cap F_2| + |F_2 \setminus F_1| \frac{1+\epsilon}{|F_2 \setminus F_1|}$$
$$= 2|F_1 \cap F_2| + 1 + \epsilon$$

Proof (cont'd)

Use fact that $\frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$ to define weights.



Greedy:

- Adds all of $F_1 \cap F_2$
- Adds all of $F_1 \setminus F_2$
- Can't add any of $F_2 \setminus F_1$

$$w(\text{greedy}) = 2|F_1 \cap F_2| + |F_1 \setminus F_2| \frac{1}{|F_1 \setminus F_2|}$$
$$= 2|F_1 \cap F_2| + 1$$

$$w(F_2) = 2|F_1 \cap F_2| + |F_2 \setminus F_1| \frac{1+\epsilon}{|F_2 \setminus F_1|}$$
$$= 2|F_1 \cap F_2| + 1 + \epsilon$$

Greedy not optimal: contradiction!